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> cs > arXiv:2412.18202

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[Submitted on 24 Dec 2024 (v1), last revised 21 Jul 2025 (this version, v6)]

Developing Cryptocurrency Trading Strategy Based on Autoencoder-CNN-GANs Algorithms

Zhuohuan Hu, Richard Yu, Zizhou Zhang, Haoran Zheng, Qianying Liu, Yining Zhou

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This paper leverages machine learning algorithms to forecast and analyze financial time series. The process begins with a denoising autoencoder to filter out random noise fluctuations from the main contract price data. Then, one-dimensional convolution reduces the dimensionality of the filtered data and extracts key information. The filtered and dimensionality-reduced price data is fed into a GANs network, and its output serve as input of a fully connected network. Through cross-validation, a model is trained to capture features that precede large price fluctuations. The model predicts the likelihood and direction of significant price changes in real-time price sequences, placing trades at moments of high prediction accuracy. Empirical results demonstrate that using autoencoders and convolution to filter and denoise financial data,

combined with GANs, achieves a certain level of predictive performance, validating the capabilities of machine learning algorithms to discover underlying patterns in financial sequences. Keywords - CNN;GANs; Cryptocurrency; Prediction.

Comments: The paper was accepted by 2024 4th International Conference on Artificial Intelligence, Robotics, and Communication(ICAIRC 2024)
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[v5] Thu, 29 May 2025 05:52:32 UTC (1,238 KB)
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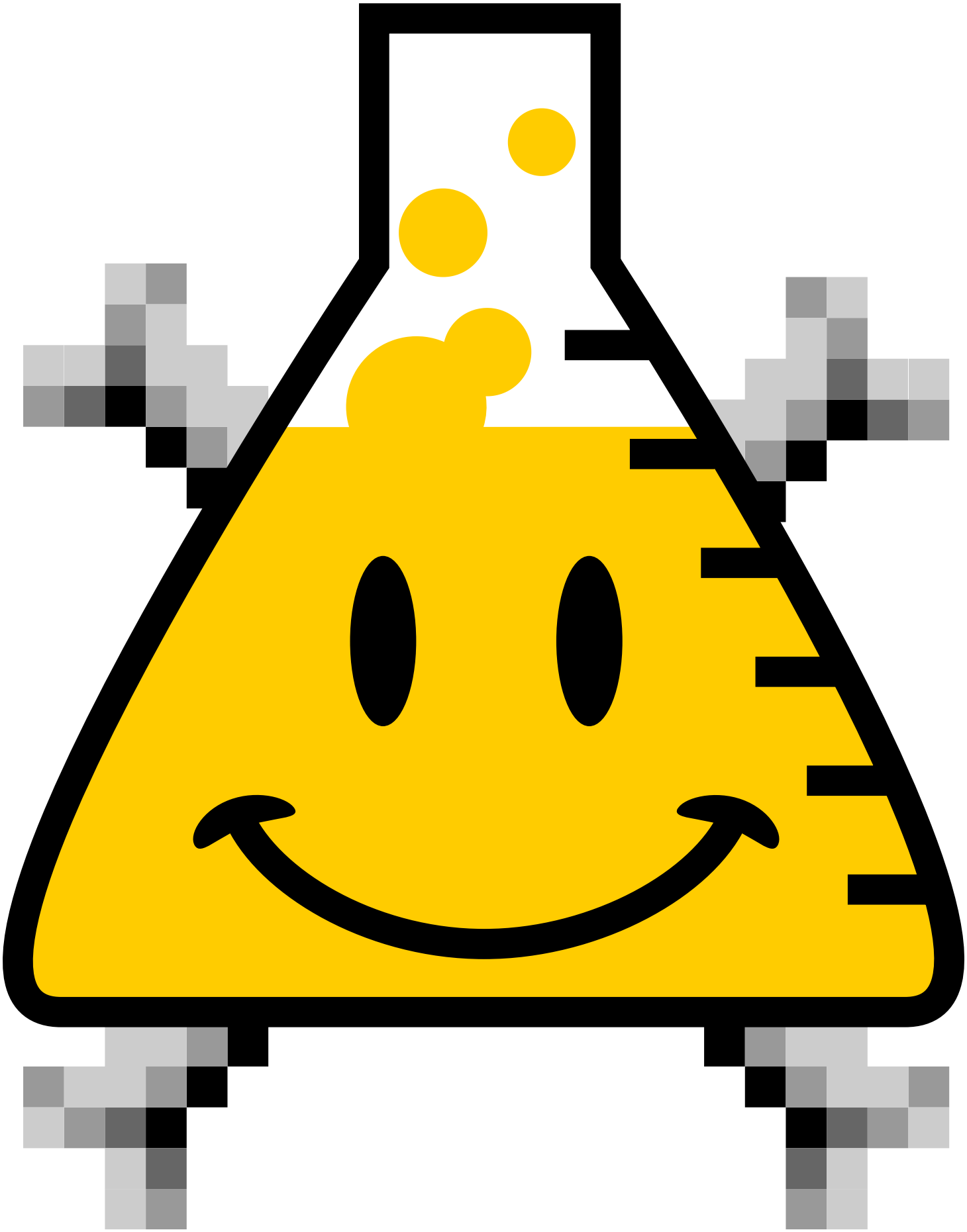

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